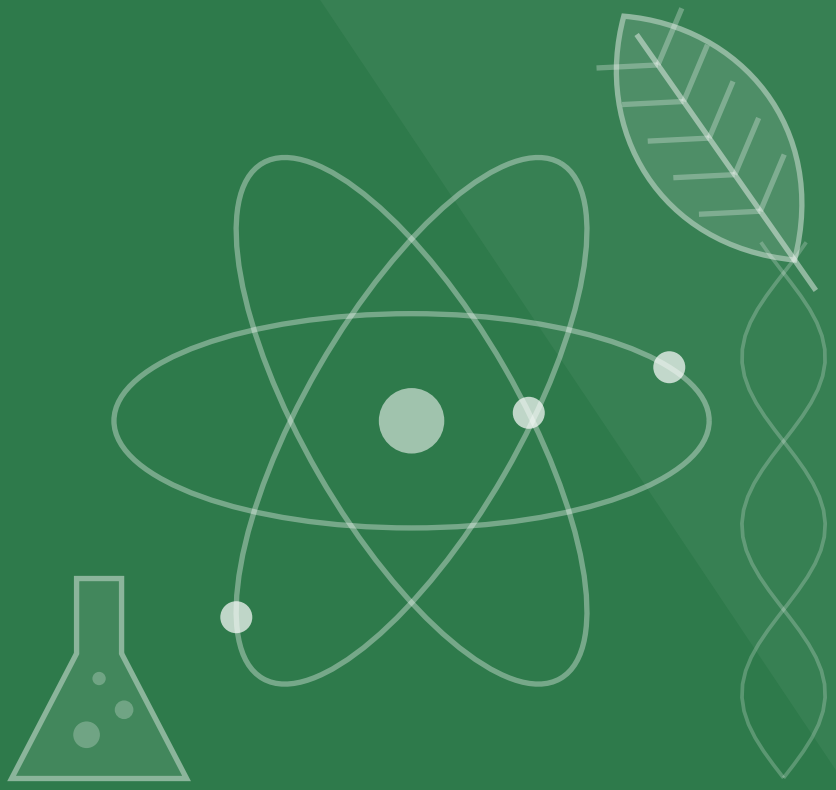
VENVLEARNING

TEACHER EDITION



Science

GRADE 4

Guided lessons, pacing, and complete answer keys

2 units

VENVLEARNING

Grade 4
Science

TEACHER EDITION

Unit 1 The Human Body

Unit 2 Energy, Heat, and Light

A VenvLearning family homeschool curriculum

How to use this Teacher Edition

This book holds everything needed to teach the course: unit overviews, a plan for every lesson with what to say and do, the student lesson with Try It answers, and answer keys for every workbook page, weekly check and unit test.

- Daily routine (about 45–55 minutes): Engage 5 · Investigate 15 · Explain 10 · Guided Practice 7 · Independent 10 · Exit Ticket 3.
- Say: lines are a suggested script. Use your own words.
- Two lessons a week, then the Weekly Check (about 15 minutes), then corrections or a project.
- Each plan's Pacing row and each unit's pacing guide give the week and lesson order. Set the days to fit your own calendar.

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UNIT 1 · TEACHER OVERVIEW

The Human Body

Thinking like a scientist; cells, tissues, and organs; and the body systems that keep you alive.

Your body is an amazing machine made of trillions of cells. In this unit you'll learn to think like a scientist: observing, asking testable questions, running fair tests, and using evidence. Then you'll use those skills to explore how your body is organized and how its systems work together.

Big ideas

- Tell observations from inferences, and plan fair tests.
- Support claims with evidence and reasoning (CER).
- Describe levels of organization from cells to systems.
- Explain the jobs of the digestive, respiratory, circulatory, skeletal, muscular, and nervous systems and how they work together.

Standards

Code	Standard
4-LS1-1	Internal and external structures support survival, growth, behavior, and reproduction.
SEP 1, 3, 4, 7	Ask questions; plan and carry out investigations; analyze data; argue from evidence.

What students should already know

- Living things need food, water, and air.

Materials for the unit

- Paper bags and small objects
- Paper towels, cups, measuring cup
- Building blocks
- Zip bags, crackers, vinegar
- Timer

Pacing guide

Teach the lessons in order, then give the week's check. Set the days to fit your own calendar.

Week	Order	Lesson	Title
Week 1	Lesson 1 of 2	1.1	Thinking Like a Scientist
Week 1	Lesson 2 of 2	1.2	Fair Tests and Evidence
Week 1	End of week	Check	Weekly Check 1: Thinking Like a Scientist
Week 2	Lesson 1 of 2	1.3	Cells, Tissues, Organs, and Systems
Week 2	Lesson 2 of 2	1.4	Body Systems Work Together
Week 2	End of week	Check	Weekly Check 2: Body Systems
Week 3	Lesson 1 of 2	1.5	The Digestive System
Week 3	Lesson 2 of 2	1.6	The Respiratory System
Week 3	End of week	Check	Weekly Check 3: Digestion and Breathing
Week 4	Lesson 1 of 2	1.7	The Heart and Blood Vessels
Week 4	Lesson 2 of 2	1.8	Bones, Muscles, and Healthy Choices
Week 4	Before the test	Review	Unit 1 Review
Week 4	End of unit	Test	Unit 1 Test

TEACHING THIS UNIT AT HOME

Take pulses and count breaths as a family before and after a walk, and compare the results.

UNIT VOCABULARY

observation	Information gathered with senses or tools.
inference	A conclusion based on observations.
fair test	Only one variable is changed.
cell	The smallest living unit.
organ system	Organs that work together.
nutrients	Substances in food the body uses.

LESSON 1.1 · TEACHER PLAN
Week 1 · Lesson 1 of 2
Unit 1: The Human Body

Lesson 1.1: Thinking Like a Scientist

Student goal	I can tell the difference between an observation and an inference, and ask a question that can be investigated.
Objective	Students distinguish observations from inferences and write testable questions.
Standard	Science and Engineering Practices 1, 3 — Ask questions; plan and carry out investigations.
Pacing	Unit 1, week 1, lesson 1 of 2
Time	About 50 minutes
Materials	Paper bags with hidden objects, workbook Lesson 1.1
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Cookie jar story: see vs think.
Investigate	15	Mystery Bag investigation.
Explain	10	Chart observations vs inferences. Model a testable question.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook lesson pages: Warm-Up through Exit Ticket.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Treating inferences as facts.

SUPPORT

- Sentence frames: I observe… / I infer…

EXTEND

- Observe a pet or bird for 10 minutes and record observations.

STUDENT LESSON 1.1 · WITH ANSWERS

Lesson 1.1: Thinking Like a Scientist

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.1 in this book with your teacher.
- [ ] Do the Investigate activity: Mystery Bag.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.1, Part A.
- [ ] Complete the rest of Workbook Lesson 1.1 (Parts B–D).

I WONDER

You walk into the kitchen and see crumbs on the counter and an empty cookie jar. What do you SEE? What do you THINK happened?

Scientists are curious people who ask questions about the world and then look for evidence to answer them. Two of their most important tools are observations and inferences.

Observation	Inference
what you notice with your senses or tools	what you figure out from what you noticed
“The cookie jar is empty.”	“Someone ate the cookies.”
“The ground is wet.”	“It rained last night.”
“The thermometer reads 38°F.”	“It is too cold to swim outside.”

An inference can be wrong. Maybe the cookies were moved to another jar! Good scientists gather more observations to test their inferences.

EXPLAIN IT · ASK A TESTABLE QUESTION

You notice that a puddle in the sun dries faster than a puddle in the shade.

1 Observation: the sunny puddle disappeared first.

2 Question that can't be tested easily: “Why does water exist?”

3 Testable question: “Does water evaporate faster in sunlight than in shade?” You can set up two cups of water and measure.

Explanation: A testable question can be answered by observing, measuring, or experimenting.

INVESTIGATE · MYSTERY BAG

What can you figure out about an object you can't see?

You need: a paper bag with a hidden object (chosen by your teacher); paper and a pencil.

1 Without looking, shake the bag and listen. Feel the object through the paper.

2 List at least four observations (sound, weight, shape, texture).

3 Write one inference about what the object is, and the observations that support it.

4 Open the bag. Was your inference correct?

Record: a two-column chart: Observations | Inferences.

★ KEY IDEA

Observations come from your senses and tools. Inferences are explanations based on observations. Testable questions can be answered with evidence.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. Observation or inference: “The leaf is green and has jagged edges.”

Answer: observation

2. Observation or inference: “The dog is hungry because it is barking at its bowl.”

Answer: inference

3. Make this question testable: “Why do plants like light?”

Answer: e.g. Does a bean plant grow taller with more hours of light?

Workbook Answer Key · Lesson 1.1

#	Question	Answer
1	Observation or inference: “The leaf is green and has jagged edges.”	observation
2	Observation or inference: “The dog is hungry because it is barking at its bowl.”	inference
3	Make this question testable: “Why do plants like light?”	e.g. Does a bean plant grow taller with more hours of light?
4	Name two of your senses.	e.g. sight, hearing, touch, smell, taste
5	Name one tool that helps scientists observe.	e.g. magnifying glass, ruler, thermometer
6	“The water is 20°C.” Observation or inference?	observation

#	Question	Answer
7	“The ice melted because the room is warm.” Observation or inference?	inference
8	“The rock feels rough.” Observation or inference?	observation
9	“A bird built this nest.” Observation or inference?	inference
10	Look out a window. Write two observations and one inference.	Answers vary; observations use senses, the inference explains them.
11	Write a testable question about something in your home.	Answers vary; must be answerable by observing or measuring.
12	In your own words, what is the difference between an observation and an inference?	An observation is what you notice with your senses or tools; an inference is what you conclude from it.

LESSON 1.2 · TEACHER PLAN
Week 1 · Lesson 2 of 2
Unit 1: The Human Body

Lesson 1.2: Fair Tests and Evidence

Student goal	I can plan a fair test and use evidence to support a claim.
Objective	Students plan fair tests and write CER.
Standard	Science and Engineering Practices 3, 4, 7 — Plan and carry out fair tests; construct explanations from evidence.
Pacing	Unit 1, week 1, lesson 2 of 2
Time	About 50 minutes
Materials	Paper towels, measuring cups, water, workbook Lesson 1.2
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Paper airplane race.
Investigate	15	Paper Towel Test.
Explain	10	Teach variables; CER.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook. The Weekly Check comes next.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Changing more than one variable.

SUPPORT

- Change/Same/Measure chart.

EXTEND

- Repeat the test three times and average the results.

STUDENT LESSON 1.2 · WITH ANSWERS

Lesson 1.2: Fair Tests and Evidence

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.2 in this book with your teacher.
- [ ] Do the Investigate activity: The Paper Towel Test.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.2, Part A.
- [ ] Complete the rest of Workbook Lesson 1.2 (Parts B–D).

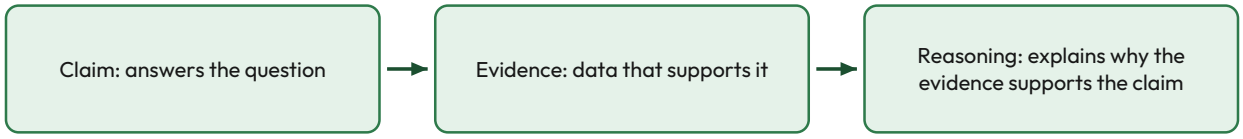
I WONDER

Two friends race paper airplanes. One throws hard and one throws gently. Is it a fair race? Why or why not?

In a fair test, you change only one thing, the variable you test, and keep everything else the same. Then you measure what happens. If you change two things at once, you can't tell which one caused the result.

Question	Change (test)	Keep the same	Measure
Which paper towel brand absorbs the most water?	brand of towel	towel size, amount of water, time	water left in the cup (mL)

After a test, scientists write a claim, support it with evidence (data), and give reasoning that explains why the evidence supports the claim. This is called CER.



INVESTIGATE · THE PAPER TOWEL TEST

Which paper towel absorbs the most water?

You need: 2 or 3 kinds of paper towel cut to the same size; measuring cup; water; 3 cups.

1. Pour 100 mL of water into each cup.
2. Dip one towel square in each cup for 10 seconds.
3. Lift the towel and let it drip back into the cup for 10 seconds.
4. Measure the water left in each cup.

Record: a data table, then a CER: Which towel absorbed the most? What is your evidence?

★ KEY IDEA

Change one variable, keep the rest the same, measure the result, and support your claim with evidence and reasoning.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. Why must the towel squares be the same size?

Answer: So that only the brand changes; size would also affect how much water is absorbed.

2. What is the claim in CER?

Answer: a statement that answers the question

3. A student tests plant growth but gives one plant more water AND more light. What is wrong?

Answer: Two variables changed, so you can't tell which caused the result.

Workbook Answer Key · Lesson 1.2

#	Question	Answer
1	Why must the towel squares be the same size?	So that only the brand changes; size would also affect how much water is absorbed.
2	What is the claim in CER?	a statement that answers the question
3	A student tests plant growth but gives one plant more water AND more light. What is wrong?	Two variables changed, so you can't tell which caused the result.
4	What does a fair test change?	only one variable
5	What do you call data that supports a claim?	evidence
6	You want to know if a ball bounces higher on carpet or wood. What will you change? What will you keep the same? What will you measure?	Change: floor surface. Same: ball, drop height. Measure: bounce height.
7	Brand A towel left 40 mL; Brand B left 65 mL; Brand C left 52 mL (from 100 mL). Which absorbed the most?	Brand A (it absorbed 60 mL)
8	Write a CER for the towel data above.	Claim: Brand A absorbs the most. Evidence: only 40 mL was left (60 mL absorbed) vs 35 and 48 mL. Reasoning: less water left means more was soaked up.
9	Why do scientists keep all variables the same except one?	So they know the one change caused the result.

WEEKLY CHECK · ANSWER KEY
After Lesson 1.2

About 15 minutes

Weekly Check 1: Thinking Like a Scientist — Answer Key

Answers

#	Question	Answer
1	Observation or inference: “The sky is cloudy.”	observation
2	Observation or inference: “It will probably rain.”	inference
3	Make this testable: “Why are some balls bouncy?”	e.g. Which ball bounces highest when dropped from 1 meter?
4	A student tests whether salt water or fresh water freezes faster, but puts one cup in the freezer and one in the refrigerator. What is wrong?	Two variables changed (type of water and location), so it isn't a fair test.
5	What does the E in CER stand for?	evidence

LESSON 1.3 · TEACHER PLAN
Week 2 · Lesson 1 of 2
Unit 1: The Human Body

Lesson 1.3: Cells, Tissues, Organs, and Systems

Student goal	I can describe how the body is organized from cells to organ systems.
Objective	Students describe levels of organization.
Standard	4-LS1-1 — Internal structures support survival and function.
Pacing	Unit 1, week 2, lesson 1 of 2
Time	About 50 minutes
Materials	Building blocks, sticky notes, workbook Lesson 1.3
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Brick → wall → room → house.
Investigate	15	Build the Levels.
Explain	10	Teach the flow chart; cell shapes fit jobs.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook lesson pages: Warm-Up through Exit Ticket.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking organs are made of only one kind of cell.

SUPPORT

- Flow chart with pictures.

EXTEND

- Look at onion skin or a leaf with a magnifier or microscope.

STUDENT LESSON 1.3 · WITH ANSWERS

Lesson 1.3: Cells, Tissues, Organs, and Systems

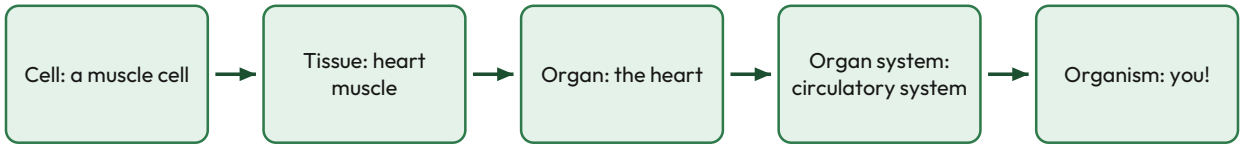
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.3 in this book with your teacher.
- [ ] Do the Investigate activity: Build the Levels.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.3, Part A.
- [ ] Complete the rest of Workbook Lesson 1.3 (Parts B–D).

I WONDER

A brick is part of a wall, a wall is part of a room, and a room is part of a house. How might your body be built the same way?

Your body is made of trillions of tiny cells, so small you need a microscope to see them. Similar cells work together as tissues, such as muscle tissue. Different tissues form organs, like the heart or stomach. Organs work together in organ systems.



There are many kinds of cells. Nerve cells are long and thin to carry messages. Red blood cells are round and flat to carry oxygen. A cell's shape fits its job.

INVESTIGATE · BUILD THE LEVELS

How can a model show levels of organization?

You need: building blocks or LEGO bricks; sticky notes.

1. Let one brick stand for one cell.
2. Join several matching bricks into a "tissue."
3. Combine different "tissues" into an "organ."
4. Put several organs together into a "system." Label each level with a sticky note.

Record: a sketch of your model with each level labeled.

★ KEY IDEA

Cells → tissues → organs → organ systems → organism.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. What is the smallest living unit of the body?

Answer: a cell

2. Put in order: organ, cell, organ system, tissue

Answer: cell, tissue, organ, organ system

3. Name one organ.

Answer: e.g. heart, lungs, stomach, brain, skin

Workbook Answer Key · Lesson 1.3

#	Question	Answer
1	What is the smallest living unit of the body?	a cell
2	Put in order: organ, cell, organ system, tissue	cell, tissue, organ, organ system
3	Name one organ.	e.g. heart, lungs, stomach, brain, skin
4	A group of similar cells is a…	tissue
5	The heart is an…	organ
6	Smallest level of organization	cell
7	Organs working together form an…	organ system
8	Why are nerve cells long and thin?	to carry messages over long distances
9	Is skin an organ?	yes (the largest organ)
10	Use the brick-and-house comparison to explain levels of organization.	Cells are like bricks; tissues like walls; organs like rooms; systems like the whole house.
11	Draw and label the four levels, from cell to organ system.	cell, tissue, organ, organ system (e.g. muscle cell, muscle tissue, heart, circulatory system)

LESSON 1.4 · TEACHER PLAN
Week 2 · Lesson 2 of 2
Unit 1: The Human Body

Lesson 1.4: Body Systems Work Together

Student goal	I can name major body systems, describe each one's main job, and explain how they work together.
Objective	Students identify body systems and their interactions.
Standard	4-LS1-1 — Internal structures function to support survival, growth, and behavior.
Pacing	Unit 1, week 2, lesson 2 of 2
Time	About 50 minutes
Materials	System index cards, workbook Lesson 1.4
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Soccer kick.
Investigate	15	Systems Relay.
Explain	10	Teach the systems chart. Model the soccer example.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook. The Weekly Check comes next.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking each system works alone.

SUPPORT

- Picture cards for each system.

EXTEND

- Make a body-systems poster.

STUDENT LESSON 1.4 · WITH ANSWERS

Lesson 1.4: Body Systems Work Together

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.4 in this book with your teacher.
- [ ] Do the Investigate activity: Systems Relay.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.4, Part A.
- [ ] Complete the rest of Workbook Lesson 1.4 (Parts B–E).

I WONDER

Think about kicking a soccer ball. Which parts of your body are working?

System	Main organs	Main job
skeletal	bones	supports, protects, works with muscles to move
muscular	muscles	moves bones and organs
nervous	brain, spinal cord, nerves	senses, thinks, sends messages
digestive	mouth, stomach, intestines	breaks food into nutrients
respiratory	nose, windpipe, lungs	takes in oxygen, gets rid of carbon dioxide
circulatory	heart, blood vessels, blood	carries oxygen and nutrients to cells

EXPLAIN IT · SYSTEMS WORKING TOGETHER

Kicking a soccer ball

1 Nervous: your eyes see the ball and your brain sends a message to your leg.
2 Muscular and skeletal: leg muscles pull on bones to swing your leg.
3 Respiratory and circulatory: you breathe faster and your heart pumps faster to deliver oxygen to your muscles.
4 Digestive: nutrients from food give your cells energy.

Explanation: No system works alone; all of them work together to keep you alive and moving.

INVESTIGATE · SYSTEMS RELAY

Which systems work when you do an everyday task?

You need: index cards labeled with the six systems; a timer.

1 Choose an action: eating an apple, running up stairs, or reading a book aloud.

2 Do the action (safely!).

3 Hold up every system card you think was working, and explain why.

Record: the action and the systems used, with one reason for each.

★ KEY IDEA

Each body system has a main job, and the systems depend on one another.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. Which system carries oxygen to cells?

Answer: circulatory

2. Which system breaks down food?

Answer: digestive

3. Which systems help you jump?

Answer: nervous, muscular, skeletal (also respiratory, circulatory)

Workbook Answer Key · Lesson 1.4

#	Question	Answer
1	Which system carries oxygen to cells?	circulatory
2	Which system breaks down food?	digestive
3	Which systems help you jump?	nervous, muscular, skeletal (also respiratory, circulatory)
4	The brain belongs to the ___ system.	nervous
5	Bones belong to the ___ system.	skeletal
6	lungs	respiratory
7	heart	circulatory
8	stomach	digestive
9	spinal cord	nervous
10	skull	skeletal
11	biceps	muscular

#	Question	Answer
12	Choose an activity (swimming, playing piano, eating lunch). Explain how three systems work together.	Answers vary; three systems with jobs.
13	Why do you breathe faster when you run?	Muscles need more oxygen; the respiratory and circulatory systems work harder to deliver it.

WEEKLY CHECK · ANSWER KEY
After Lesson 1.4

About 15 minutes

Weekly Check 2: Body Systems — Answer Key

Answers

#	Question	Answer
1	Put in order: organ, cell, tissue, organ system	cell, tissue, organ, organ system
2	Which system includes the brain?	nervous
3	Which system breaks down food?	digestive
4	Which system moves blood?	circulatory
5	Name two systems that work together when you climb stairs and explain how.	e.g. muscular and skeletal: muscles pull bones to lift legs.

LESSON 1.5 · TEACHER PLAN
Week 3 · Lesson 1 of 2
Unit 1: The Human Body

Lesson 1.5: The Digestive System

Student goal	I can describe the path of food through the digestive system and explain what each organ does.
Objective	Students trace food through the digestive system.
Standard	4-LS1-1 — Internal structures support survival.
Pacing	Unit 1, week 3, lesson 1 of 2
Time	About 50 minutes
Materials	Zip bags, crackers, water, vinegar, workbook Lesson 1.5
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Where does breakfast go?
Investigate	15	Stomach in a Bag.
Explain	10	Teach the path; mechanical vs chemical digestion.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook lesson pages: Warm-Up through Exit Ticket.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking food goes straight to the stomach and stops.

SUPPORT

- Trace the path on a body outline.

EXTEND

- Measure 20 feet of string to see the small intestine's length.

STUDENT LESSON 1.5 · WITH ANSWERS

Lesson 1.5: The Digestive System

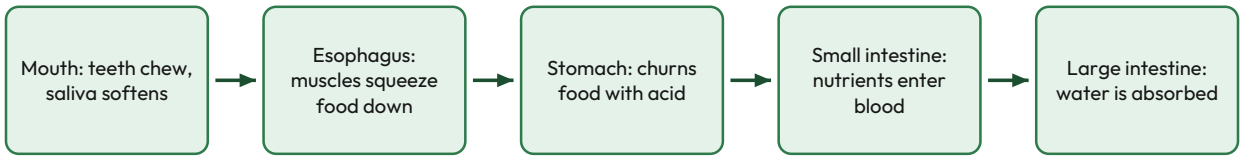
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.5 in this book with your teacher.
- [ ] Do the Investigate activity: Stomach in a Bag.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.5, Part A.
- [ ] Complete the rest of Workbook Lesson 1.5 (Parts B–D).

I WONDER

Where does your breakfast go after you swallow it?

Your body can't use a sandwich as a sandwich. The digestive system breaks food into tiny nutrients that can pass into your blood and travel to your cells.



Digestion is both mechanical (chewing and churning break food into smaller pieces) and chemical (saliva, stomach acid, and other juices break food down into nutrients). The liver and pancreas make juices that help in the small intestine. In an adult, the small intestine is about 20 feet long, coiled up to fit inside the body!

INVESTIGATE · STOMACH IN A BAG

How does the stomach help break down food?

You need: zip-top plastic bag; 4 crackers; 1/2 cup water; a few drops of vinegar (to stand for acid).

1. Put the crackers in the bag. Observe them.
2. Add the water and vinegar and seal the bag tightly.
3. Gently squeeze and mash the bag for 2 minutes, the way stomach muscles churn.
4. Observe the crackers again.

Record: before-and-after observations, and which part of digestion the squeezing models.

Safety: Do not eat or drink anything used in an investigation.

★ KEY IDEA

Food travels mouth → esophagus → stomach → small intestine → large intestine. Nutrients enter the blood in the small intestine.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. Where do most nutrients enter the blood?

Answer: the small intestine

2. What does the large intestine absorb?

Answer: water

3. What did squeezing the bag model?

Answer: the stomach churning food (mechanical digestion)

Workbook Answer Key · Lesson 1.5

#	Question	Answer
1	Where do most nutrients enter the blood?	the small intestine
2	What does the large intestine absorb?	water
3	What did squeezing the bag model?	the stomach churning food (mechanical digestion)
4	What do teeth do to food?	chew/break it into smaller pieces
5	What are nutrients?	substances in food the body uses
6	Put in order: stomach, mouth, large intestine, esophagus, small intestine	mouth, esophagus, stomach, small intestine, large intestine
7	Mechanical or chemical: teeth chewing	mechanical
8	Mechanical or chemical: stomach acid	chemical
9	Which organ is a long, coiled tube where nutrients enter the blood? A. esophagus B. stomach C. small intestine D. large intestine	C. small intestine
10	Explain why food must be broken down before the body can use it.	Cells can only use tiny nutrients that pass into the blood, not whole pieces of food.
11	Describe what happened to the crackers in the bag and what it shows about digestion.	They became mushy and broke apart; the stomach mixes and breaks down food.

LESSON 1.6 · TEACHER PLAN
Week 3 · Lesson 2 of 2
Unit 1: The Human Body

Lesson 1.6: The Respiratory System

Student goal	I can explain how the respiratory system brings oxygen into the body and removes carbon dioxide.
Objective	Students explain gas exchange and breathing.
Standard	4-LS1-1 — Internal structures support survival.
Pacing	Unit 1, week 3, lesson 2 of 2
Time	About 50 minutes
Materials	Timer, workbook Lesson 1.6
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Hold your breath.
Investigate	15	Breathing Rate investigation.
Explain	10	Teach the path and the diaphragm.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook. The Weekly Check comes next.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking we breathe out no oxygen at all, or that lungs are muscles.

SUPPORT

- Hands on ribs while breathing.

EXTEND

- Build a balloon lung model in a plastic bottle.

STUDENT LESSON 1.6 · WITH ANSWERS

Lesson 1.6: The Respiratory System

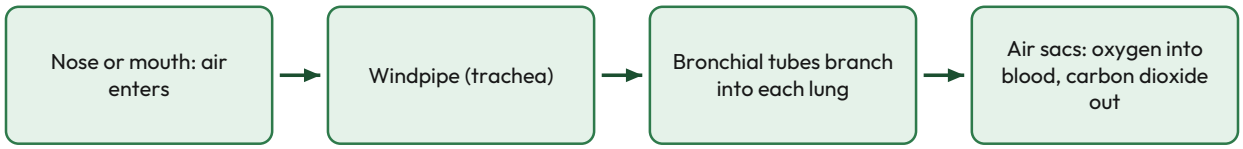
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.6 in this book with your teacher.
- [ ] Do the Investigate activity: Breathing Rate.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.6, Part A.
- [ ] Complete the rest of Workbook Lesson 1.6 (Parts B–E).

I WONDER

Hold your breath for a few seconds. Why can't you hold it forever?

Every cell in your body needs oxygen to release energy from nutrients. Cells also make carbon dioxide, a waste gas. The respiratory system brings oxygen in and sends carbon dioxide out.



When the diaphragm tightens and moves down, the chest gets bigger and air rushes into the lungs. When it relaxes and moves up, air is pushed out. Inside the lungs are millions of tiny air sacs wrapped in blood vessels, where oxygen moves into the blood and carbon dioxide moves out.

INVESTIGATE · BREATHING RATE

How does exercise change how fast you breathe?

You need: a timer; a partner.

1. Sit quietly for 1 minute. Count your breaths for 30 seconds and multiply by 2.
2. Do jumping jacks for 1 minute.
3. Right away, count your breaths for 30 seconds and multiply by 2.
4. Rest 3 minutes and count again.

Record: breaths per minute at rest, after exercise, and after resting, in a table and a bar graph.

★ KEY IDEA

You breathe in oxygen that cells need and breathe out carbon dioxide that cells make. Gases are exchanged in the air sacs.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. What gas do your cells need?

Answer: oxygen

2. What muscle helps you breathe?

Answer: the diaphragm

3. Why does breathing speed up during exercise?

Answer: Muscles need more oxygen and make more carbon dioxide.

Workbook Answer Key · Lesson 1.6

#	Question	Answer
1	What gas do your cells need?	oxygen
2	What muscle helps you breathe?	the diaphragm
3	Why does breathing speed up during exercise?	Muscles need more oxygen and make more carbon dioxide.
4	The gas you breathe out is...	carbon dioxide
5	The windpipe is also called the...	trachea
6	Order: air sacs, nose, windpipe, bronchial tubes	nose, windpipe, bronchial tubes, air sacs
7	Where does oxygen pass into the blood?	in the air sacs of the lungs
8	Maya breathed 18 times per minute at rest and 36 after exercise. How many more?	18 more (twice as many)
9	Explain how the respiratory and circulatory systems work together.	The lungs put oxygen into the blood; the heart pumps that blood to cells and brings carbon dioxide back to the lungs.
10	Make a bar graph of your breathing data: at rest, after exercise, after resting.	Bars labeled with breaths per minute; after-exercise bar is tallest.
11	What happens when the diaphragm moves down?	The chest gets bigger and air flows into the lungs.

WEEKLY CHECK · ANSWER KEY
After Lesson 1.6

About 15 minutes

Weekly Check 3: Digestion and Breathing — Answer Key

Answers

#	Question	Answer
1	Where do most nutrients enter the blood?	small intestine
2	What does the large intestine absorb?	water
3	What gas do you breathe out?	carbon dioxide
4	Where are gases exchanged in the lungs?	air sacs
5	Which is mechanical digestion? A. stomach acid B. chewing C. saliva chemicals	B. chewing
6	Ben breathes 16 times a minute at rest and 34 after running. Explain why.	Running muscles need more oxygen and produce more carbon dioxide, so he breathes faster.

LESSON 1.7 · TEACHER PLAN
Week 4 · Lesson 1 of 2
Unit 1: The Human Body

Lesson 1.7: The Heart and Blood Vessels

Student goal	I can describe how the heart and blood vessels move blood through the body, and measure my pulse.
Objective	Students describe circulation and measure pulse.
Standard	4-LS1-1 — Internal structures support survival.
Pacing	Unit 1, week 4, lesson 1 of 2
Time	About 50 minutes
Materials	Timer, workbook Lesson 1.7
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Fist-size heart.
Investigate	15	Take Your Pulse.
Explain	10	Teach vessels; the cycle diagram.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook lesson pages: Warm-Up through Exit Ticket.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking blood in veins is blue.

SUPPORT

- Body outline with red and blue arrows labeled “to heart” / “from heart.”

EXTEND

- Calculate how many times your heart beats in a day.

STUDENT LESSON 1.7 · WITH ANSWERS

Lesson 1.7: The Heart and Blood Vessels

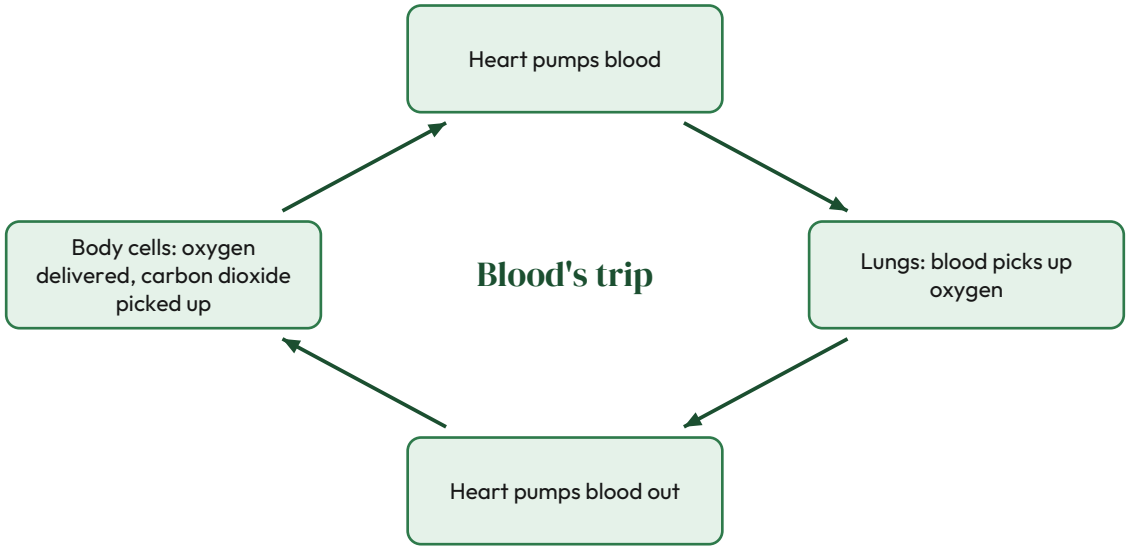
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.7 in this book with your teacher.
- [ ] Do the Investigate activity: Take Your Pulse.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.7, Part A.
- [ ] Complete the rest of Workbook Lesson 1.7 (Parts B–D).

I WONDER

Make a fist. Your heart is about that size. How many times do you think it beats in one minute?

The heart is a muscle about the size of your fist. It has four chambers and pumps blood through a huge network of blood vessels. Arteries carry blood away from the heart. Capillaries are tiny vessels where oxygen and nutrients move into cells and wastes move out. Veins carry blood back to the heart.



INVESTIGATE · TAKE YOUR PULSE

How does activity change heart rate?

You need: a timer.

1. Place two fingers (not your thumb) on the side of your neck or the inside of your wrist.
2. Count the beats for 15 seconds. Multiply by 4 to find beats per minute.
3. Run in place for 1 minute, then measure again.

4 Rest 3 minutes and measure again.

Record: beats per minute in a table; compare with your breathing data from Lesson 1.6.

! WATCH OUT

Blood in veins is not blue. Blood is always red; veins can look bluish through skin because of how light passes through it.

★ KEY IDEA

The heart pumps blood through arteries to capillaries and back through veins, delivering oxygen and nutrients and carrying away wastes.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. Which vessels carry blood away from the heart?

Answer: arteries

2. Where do oxygen and nutrients move into cells?

Answer: capillaries

3. 15-second count is 22 beats. Beats per minute?

Answer: 88

Workbook Answer Key · Lesson 1.7

#	Question	Answer
1	Which vessels carry blood away from the heart?	arteries
2	Where do oxygen and nutrients move into cells?	capillaries
3	15-second count is 22 beats. Beats per minute?	88
4	The heart is about the size of your…	fist
5	Veins carry blood…	back to the heart
6	Pulse count of 20 in 15 seconds = ___ beats per minute	80
7	Pulse count of 30 in 15 seconds = ___ beats per minute	120
8	Why does heart rate go up during exercise?	Muscles need more oxygen and nutrients, so blood must move faster.
9	Which vessel is the tiniest? A. artery B. vein C. capillary	C. capillary

#	Question	Answer
10	Trace a drop of blood from the heart to the lungs and back to the body.	Heart → lungs (pick up oxygen) → heart → body cells (deliver oxygen) → heart.
11	Why do the respiratory and circulatory systems both speed up during exercise?	Muscles need more oxygen; lungs take in more and the heart delivers it faster.

LESSON 1.8 · TEACHER PLAN
Week 4 · Lesson 2 of 2
Unit 1: The Human Body

Lesson 1.8: Bones, Muscles, and Healthy Choices

Student goal	I can explain how bones and muscles work together and describe choices that keep body systems healthy.
Objective	Students explain bone-muscle teamwork and healthy habits.
Standard	4-LS1-1 — Internal structures support survival, growth, and behavior.
Pacing	Unit 1, week 4, lesson 2 of 2
Time	About 50 minutes
Materials	Workbook Lesson 1.8, Unit 1 Review
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Bend your arm.
Investigate	15	Muscle Partners.
Explain	10	Teach bones, tendons, muscle pairs. Healthy choices chart.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook; then the Unit 1 Review.
Exit Ticket	3	Next: Unit 1 Test.

COMMON MISCONCEPTIONS

- Thinking muscles push.

SUPPORT

- Model with two rubber bands and a hinged ruler.

EXTEND

- Track sleep and activity for a week.

STUDENT LESSON 1.8 · WITH ANSWERS

Lesson 1.8: Bones, Muscles, and Healthy Choices

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.8 in this book with your teacher.
- [ ] Do the Investigate activity: Muscle Partners.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.8, Part A.
- [ ] Complete the rest of Workbook Lesson 1.8 (Parts B–E).

I WONDER

Bend and straighten your arm. What do you feel happening in your upper arm?

An adult skeleton has 206 bones. Bones support the body, protect organs (the skull protects the brain; the ribs protect the heart and lungs), and make blood cells inside. Muscles are attached to bones by tendons. Muscles can only pull, not push, so they work in pairs.

EXPLAIN IT · MUSCLE PAIRS

Bending and straightening your arm

1 To bend: the biceps (front of upper arm) tightens and pulls the forearm up; the triceps relaxes.

2 To straighten: the triceps (back of upper arm) tightens and pulls; the biceps relaxes.

Explanation: Muscles work in pairs because each muscle can only pull.

Healthy choice	Systems it helps
active play about 60 minutes a day	muscular, skeletal, circulatory, respiratory
foods like fruits, vegetables, whole grains, and protein	digestive (and every system that needs nutrients)
9–12 hours of sleep a night for children your age	nervous (and all systems repair during sleep)
drinking water	digestive, circulatory
wearing a helmet and seat belt	skeletal, nervous

INVESTIGATE · MUSCLE PARTNERS

How do muscles work in pairs?

You need: your own arm.

1 Place your hand on your biceps. Slowly bend your arm. What does the muscle do?

2 Move your hand to your triceps. Straighten your arm. What does that muscle do?

Record: a labeled sketch of the upper arm showing which muscle pulls to bend and which pulls to straighten.

★ KEY IDEA

Bones support and protect; muscles pull in pairs to move bones at joints. Healthy habits keep all systems working.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. How many bones in an adult skeleton?

Answer: 206

2. What connects muscle to bone?

Answer: a tendon

3. Why do muscles work in pairs?

Answer: Muscles can only pull, so one pulls to bend and the other pulls to straighten.

Workbook Answer Key · Lesson 1.8

#	Question	Answer
1	How many bones in an adult skeleton?	206
2	What connects muscle to bone?	a tendon
3	Why do muscles work in pairs?	Muscles can only pull, so one pulls to bend and the other pulls to straighten.
4	What do the ribs protect?	the heart and lungs
5	A place where bones meet is a...	joint
6	Muscle that bends the arm	biceps
7	Muscle that straightens the arm	triceps
8	Connects bone to bone	ligament
9	The skull protects the...	brain

#	Question	Answer
10	Write a one-day plan with three healthy choices. Name the system each one helps.	Answers vary.
11	Describe how the skeletal and muscular systems work together to help you walk.	Muscles pull on bones at joints; bones support the body and act as levers.

UNIT REVIEW · ANSWER KEY
Study before the unit test

About 40 minutes

Unit 1 Review — Answer Key

Homework or a study session before the test.

Answers

#	Question	Answer
1	Observation or inference: “The cat is sleepy.”	inference
2	In a fair test, how many variables do you change on purpose?	one
3	Smallest unit of life	cell
4	Carries blood back to the heart	vein
5	Tiny vessels where materials pass to cells	capillaries
6	Muscle that bends the arm	biceps
7	Tube from mouth to stomach	esophagus
8	Muscle under the lungs	diaphragm
9	How do the digestive, respiratory, and circulatory systems work together to give a muscle cell energy?	Digestive gives nutrients, respiratory gives oxygen, and the circulatory system carries both to the cell.

UNIT TEST · ANSWER KEY
At the end of the unit

About 45 minutes

Unit 1 Test — Answer Key

Answers

#	Question	Answer
1	Observation or inference: “The soup is steaming.”	observation
2	Observation or inference: “The soup is too hot to eat.”	inference
3	Plan a fair test: Does a ball roll farther on carpet or on tile? Name what you change, keep the same, and measure.	Change: surface. Same: ball, ramp height, push. Measure: distance rolled.
4	Which list is in order from smallest to largest? A. organ, tissue, cell, system B. cell, tissue, organ, system C. tissue, cell, system, organ	B. cell, tissue, organ, system
5	Which system includes the lungs?	respiratory
6	Which system includes the brain and nerves?	nervous
7	Where do most nutrients pass into the blood?	small intestine
8	What gas do cells need?	oxygen
9	Which blood vessels carry blood away from the heart?	arteries
10	What connects a muscle to a bone?	tendon
11	Why do muscles work in pairs?	Muscles can only pull.
12	Which is chemical digestion? A. chewing B. stomach churning C. stomach acid breaking down food	C. stomach acid breaking down food
13	A pulse of 24 beats in 15 seconds is how many beats per minute?	96
14	Resting breaths: 18/min; after exercise: 38/min. What changed and by how much?	It increased by 20 breaths per minute.
15	Write a CER: Does exercise increase heart rate? Use this data: resting 80, after running 128 beats per minute.	Claim: yes. Evidence: 80 → 128 bpm. Reasoning: muscles need more oxygen, so the heart pumps faster.
16	Choose an activity. Explain how at least three body systems work together during it.	Rubric: three correct systems (1), a correct job for each (1), a clear explanation of how they connect (1).

Scoring

Score	What it means / next step
18–21 (85%+)	Mastered.
15–17	Proficient; review missed skills.
Below 15	Reteach and retest.

UNIT 2 · TEACHER OVERVIEW

Energy, Heat, and Light

Forms of energy, heat and temperature, conduction, convection, and radiation, conductors and insulators, how light travels, reflection, refraction, and color.

Energy is everywhere: in a rolling ball, a warm mug, a sunny window, and a rainbow. In this unit you'll discover what energy is, measure temperature, watch heat move in three different ways, and design a way to keep ice from melting. Then you'll explore light: how it travels, how we see, why shadows form, and how a glass of water can make a rainbow.

Big ideas

- Describe forms of energy and relate speed to energy.
- Measure temperature and explain how heat moves.
- Compare conductors and insulators and design a solution.
- Model how light travels, reflects, refracts, and lets us see.

Standards

Code	Standard
4-PS3-1, 2, 4	Energy, speed, and energy transfer; design a device that converts energy.
4-PS4-2	Light reflecting from objects into the eye.
3-5-ETS1-1	Define a design problem with criteria and constraints.

What students should already know

- Observations, fair tests, and data tables (Unit 1).

Materials for the unit

- Marbles, a ramp, paper cups
- Thermometers and a timer
- Food coloring, ice, a clear dish
- Insulating materials (cloth, foam, cotton)

- Flashlights, small mirrors, wax paper, foil
- A cardboard carton for a periscope

Pacing guide

Teach the lessons in order, then give the week's check. Set the days to fit your own calendar.

Week	Order	Lesson	Title
Week 1	Lesson 1 of 2	2.1	What Is Energy?
Week 1	Lesson 2 of 2	2.2	Heat and Temperature
Week 1	End of week	Check	Weekly Check 1: Energy and Heat
Week 2	Lesson 1 of 2	2.3	How Heat Moves
Week 2	Lesson 2 of 2	2.4	Conductors and Insulators
Week 2	End of week	Check	Weekly Check 2: How Heat Moves
Week 3	Lesson 1 of 2	2.5	Light Travels and How We See
Week 3	Lesson 2 of 2	2.6	Reflection, Absorption, and Transparency
Week 3	End of week	Check	Weekly Check 3: Light
Week 4	Lesson 1 of 2	2.7	Refraction and Color
Week 4	Lesson 2 of 2	2.8	Light Investigation: Claim, Evidence, and Reasoning
Week 4	Before the test	Review	Unit 2 Review
Week 4	End of unit	Test	Unit 2 Test

TEACHING THIS UNIT AT HOME

Go on a heat hunt at home: find an example of conduction, convection, and radiation in your kitchen, and test which cup keeps cocoa warm the longest.

UNIT VOCABULARY

energy	The ability to make something move or change.
heat	Energy flowing from warmer to cooler.
conductor	Lets heat move easily.
insulator	Slows heat.
reflect	Bounce off.
refraction	The bending of light.

LESSON 2.1 · TEACHER PLAN
Week 1 · Lesson 1 of 2
Unit 2: Energy, Heat, and Light

Lesson 2.1: What Is Energy?

Student goal	I can describe forms of energy and explain that faster objects have more energy.
Objective	Students describe forms of energy and relate speed to energy.
Standard	4-PS3-1, 4-PS3-2 — Use evidence to explain the relationship between speed and energy (4-PS3-1).
Pacing	Unit 2, week 1, lesson 1 of 2
Time	About 50 minutes
Materials	Ramp, books, marble, paper cup, measuring tape, workbook Lesson 2.1
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Slow vs fast bowling ball.
Investigate	15	Ramp and Cup investigation.
Explain	10	Chart forms of energy. Connect ramp height, speed, and distance.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook lesson pages: Warm-Up through Exit Ticket.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking heavier always means more energy, ignoring speed.

SUPPORT

- Picture cards for forms of energy.

EXTEND

- Try a heavier ball on the same ramp and compare.

STUDENT LESSON 2.1 · WITH ANSWERS

Lesson 2.1: What Is Energy?

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.1 in this book with your teacher.
- [ ] Do the Investigate activity: Ramp and Cup.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 2.1, Part A.
- [ ] Complete the rest of Workbook Lesson 2.1 (Parts B–E).

I WONDER

A bowling ball rolls slowly into the pins. Then it rolls fast. Which time do more pins fall? Why?

Energy is the ability to make something move or change. You can't hold energy in your hand, but you can see what it does: a ball rolls, a lamp glows, a pot of soup gets hot, a drum makes sound.

Form of energy	What it does	Example
motion	makes things move	a rolling ball, wind, a running dog
sound	travels as vibrations	a drum, a voice, thunder
light	lets us see	the sun, a flashlight, a candle
heat (thermal)	makes things warmer	a stove, a campfire, your hands rubbing together
electrical	flows through wires	a battery, an outlet, lightning

Energy can transfer, or move, from one object to another. When a moving ball hits another ball, some of its motion energy moves into the second ball. Energy can also change form: a flashlight changes electrical energy into light and a little heat.

The faster an object moves, the more motion energy it has. A fast ball hits harder, makes a louder sound, and pushes other objects farther than a slow ball of the same size.

EXPLAIN IT · USE EVIDENCE ABOUT SPEED AND ENERGY

A marble rolled down a steep ramp knocks a cup 30 cm. The same marble rolled down a low ramp knocks it 10 cm.

1 The steep ramp made the marble move faster.

2 The faster marble pushed the cup farther.

3 A bigger push means more energy was transferred.

Explanation: The faster marble had more motion energy.

INVESTIGATE · RAMP AND CUP

Does a faster marble have more energy?

You need: a ruler with a groove (or a cardboard ramp); a stack of books; a marble; a paper cup with a doorway cut out; a measuring tape.

1 Set the ramp on one book and place the cup at the bottom.

2 Let the marble go (don't push it) and measure how far the cup moves.

3 Repeat with two books, then three books.

4 Do three trials at each height and record the distances.

Record: a data table of ramp height and distance, and a sentence that uses your data to answer the question.

★ KEY IDEA

Energy makes things move or change. It comes in forms such as motion, sound, light, heat, and electrical, and it can transfer between objects. Faster objects have more motion energy.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. Name three forms of energy.

Answer: any three of motion, sound, light, heat, electrical

2. Which has more motion energy: a slow bike or a fast bike?

Answer: a fast bike

3. What energy changes happen in a flashlight?

Answer: electrical energy changes into light and heat

Workbook Answer Key · Lesson 2.1

#	Question	Answer
1	Name three forms of energy.	any three of motion, sound, light, heat, electrical
2	Which has more motion energy: a slow bike or a fast bike?	a fast bike
3	What energy changes happen in a flashlight?	electrical energy changes into light and heat
4	The ability to make something move or change	energy

#	Question	Answer
5	To move from one object to another	transfer
6	a ringing bell	sound
7	a toaster's glowing wires	heat and light
8	a kicked soccer ball	motion
9	a phone charger	electrical
10	Use your Ramp and Cup data. What happened as the ramp got higher? What does that tell you about speed and energy?	Higher ramp, faster marble, cup moved farther; faster objects have more energy.
11	Explain why a fast-moving ball hurts more when it hits you than a slow one.	The faster ball has more motion energy, so it transfers more energy to you.

LESSON 2.2 · TEACHER PLAN
Week 1 · Lesson 2 of 2
Unit 2: Energy, Heat, and Light

Lesson 2.2: Heat and Temperature

Student goal	I can measure temperature and explain that heat flows from warmer objects to cooler ones.
Objective	Students measure temperature and describe the direction of heat flow.
Standard	4-PS3-2 — Make observations to provide evidence that energy can be transferred by heat (4-PS3-2).
Pacing	Unit 2, week 1, lesson 2 of 2
Time	About 50 minutes
Materials	Cups, thermometers, timer, workbook Lesson 2.2
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Cocoa and ice water.
Investigate	15	Cooling Cups investigation.
Explain	10	Practice reading a thermometer. Draw heat-flow arrows.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook. The Weekly Check comes next.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Believing “cold” flows into warm objects.

SUPPORT

- Large thermometer model for reading practice.

EXTEND

- Graph your Cooling Cups data on a line graph.

STUDENT LESSON 2.2 · WITH ANSWERS

Lesson 2.2: Heat and Temperature

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.2 in this book with your teacher.
- [ ] Do the Investigate activity: Cooling Cups.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 2.2, Part A.
- [ ] Complete the rest of Workbook Lesson 2.2 (Parts B–E).

I WONDER

Why does a cup of hot cocoa get cold if you forget about it, but a glass of ice water gets warm?

Temperature tells how hot or cold something is. We measure it with a thermometer in degrees. In the United States we often use degrees Fahrenheit (°F); scientists use degrees Celsius (°C). Water freezes at 32°F (0°C) and boils at 212°F (100°C).

Heat is energy that moves from a warmer object to a cooler one, always in that direction. Your hot cocoa loses heat to the cooler air around it until they are the same temperature. Ice water gains heat from the warmer air. When you hold an ice cube, heat flows from your warm hand into the ice, which is why your hand feels cold.

Temperature	°F	°C
water freezes	32°F	0°C
a cool fall day	50°F	10°C
room temperature	about 70°F	about 21°C
normal body temperature	about 98.6°F	about 37°C
water boils	212°F	100°C

EXPLAIN IT · READ A THERMOMETER

The red liquid in a thermometer stops at the line halfway between 60 and 70 on the °F side.

1 Find the two labeled numbers on each side of the top of the liquid: 60 and 70.

2 The liquid is halfway between them.

3 Halfway between 60 and 70 is 65.

Explanation: The temperature is 65°F.

INVESTIGATE · COOLING CUPS

How does the temperature of warm water change over time?

You need: two cups; warm tap water and cold tap water; a thermometer; a timer.

1 Fill one cup with warm tap water and one with cold tap water.

2 Measure and record each temperature.

3 Measure both cups again every 5 minutes for 20 minutes.

4 Measure the room's air temperature too.

Record: a data table and a sentence explaining which way heat flowed for each cup.

Safety: Use warm tap water, not hot or boiling water.

! WATCH OUT

Cold is not something that flows into your hand. Heat flows out of your hand into the cold object. Heat always moves from warmer to cooler.

★ KEY IDEA

Temperature is how hot or cold something is, measured with a thermometer. Heat is energy that always flows from warmer objects to cooler ones.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. What tool measures temperature?

Answer: a thermometer

2. At what temperature does water freeze in °F?

Answer: 32°F

3. You hold a snowball. Which way does heat flow?

Answer: from your hand into the snowball

Workbook Answer Key · Lesson 2.2

#	Question	Answer
1	What tool measures temperature?	a thermometer
2	At what temperature does water freeze in °F?	32°F
3	You hold a snowball. Which way does heat flow?	from your hand into the snowball
4	How hot or cold something is	temperature

#	Question	Answer
5	Water boils at… (°C)	100°C
6	a hot pan cooling on the stove	from the pan to the air
7	a cold drink on a summer day	from the air to the drink
8	your feet on a cold tile floor	from your feet to the floor
9	an ice cube in your hand	from your hand to the ice
10	Use your Cooling Cups data. What happened to each cup's temperature? Where did each cup end up compared with the room?	The warm cup cooled and the cold cup warmed; both moved toward room temperature.
11	Explain why hot cocoa gets cold if you leave it on the table.	Heat flows from the warmer cocoa into the cooler air until they are the same temperature.

WEEKLY CHECK · ANSWER KEY
After Lesson 2.2

About 15 minutes

Weekly Check 1: Energy and Heat — Answer Key

Answers

#	Question	Answer
1	The ability to make something move or change	energy
2	Which has more motion energy: a fast car or a slow car?	a fast car
3	A tool that measures temperature	a thermometer
4	Water freezes at… (°F)	32°F
5	You hold an ice cube. Which way does heat flow?	from your hand into the ice
6	Name two forms of energy and give an example of each.	e.g. light: a lamp; sound: a drum.

LESSON 2.3 · TEACHER PLAN
Week 2 · Lesson 1 of 2
Unit 2: Energy, Heat, and Light

Lesson 2.3: How Heat Moves

Student goal	I can describe three ways heat moves: conduction, convection, and radiation.
Objective	Students identify conduction, convection, and radiation.
Standard	4-PS3-2 — Provide evidence that energy can be transferred by heat (4-PS3-2).
Pacing	Unit 2, week 2, lesson 1 of 2
Time	About 50 minutes
Materials	Clear dish, food coloring, ice, warm water, workbook Lesson 2.3
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Campfire on your face.
Investigate	15	Food Coloring Currents.
Explain	10	Chart the three ways heat moves. Model each with a gesture: touch, swirl, wave.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook lesson pages: Warm-Up through Exit Ticket.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking heat moves only by touching.

SUPPORT

- Hand motions for each kind of heat transfer.

EXTEND

- Find all three ways heat moves in your kitchen.

STUDENT LESSON 2.3 · WITH ANSWERS

Lesson 2.3: How Heat Moves

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.3 in this book with your teacher.
- [ ] Do the Investigate activity: Food Coloring Currents.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 2.3, Part A.
- [ ] Complete the rest of Workbook Lesson 2.3 (Parts B–D).

I WONDER

You sit near a campfire. Your face feels hot even though you aren't touching the fire. How does the heat reach you?

Heat moves in three ways. In conduction, heat moves through objects that are touching. A metal spoon in hot soup gets hot all the way up the handle as heat passes from particle to particle.

In convection, heat moves when a liquid or gas flows. Warm air and warm water rise, and cooler air and water sink, making a loop called a convection current. That's why the upstairs of a house is often warmer.

In radiation, heat travels as invisible waves, even through empty space. The sun's heat travels about 93 million miles through space to warm Earth. You feel radiation from a campfire or a toaster without touching it.

Way heat moves	How	Example
conduction	through touching objects	a spoon in hot soup; a pan on a stove burner
convection	by flowing liquid or gas	warm air rising from a heater; boiling water churning
radiation	as waves through space or air	sunshine on your skin; heat from a campfire

INVESTIGATE · FOOD COLORING CURRENTS

Can you see convection in water?

You need: a clear glass dish or jar of room-temperature water; a small cup of warm water with red food coloring; an ice cube made with blue food coloring.

1 Let the water in the dish sit until it is still.

2 Gently set the blue ice cube at one end.

3 Carefully pour a little red warm water in at the other end, near the bottom.

4 Watch from the side for 5 minutes. Where does each color go?

Record: a side-view drawing with arrows showing where the warm and cold water moved.

Safety: Use warm tap water only.

★ KEY IDEA

Heat moves by conduction (touching), convection (flowing liquids and gases), and radiation (waves through space).

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. How does the sun's heat reach Earth?

Answer: radiation

2. A pan gets hot on a burner. Which way does the heat move?

Answer: conduction

3. Why is the top floor of a house often warmer?

Answer: Warm air rises (convection).

Workbook Answer Key · Lesson 2.3

#	Question	Answer
1	How does the sun's heat reach Earth?	radiation
2	A pan gets hot on a burner. Which way does the heat move?	conduction
3	Why is the top floor of a house often warmer?	Warm air rises (convection).
4	Heat moving through touching objects	conduction
5	Heat traveling as waves	radiation
6	warming your hands near a fireplace	radiation
7	a metal slide getting hot in the sun and burning your legs	conduction (sun to slide is radiation)
8	warm air from a vent rising to the ceiling	convection
9	an ice pack cooling a bump	conduction
10	soup bubbling and churning in a pot	convection
11	sunlight warming a car seat	radiation

#	Question	Answer
12	Describe the convection you saw in the Food Coloring Currents investigation.	The warm red water rose; the cold blue water sank; they moved in a loop.

LESSON 2.4 · TEACHER PLAN
Week 2 · Lesson 2 of 2
Unit 2: Energy, Heat, and Light

Lesson 2.4: Conductors and Insulators

Student goal	I can test materials and design a way to slow the flow of heat.
Objective	Students compare conductors and insulators and design a heat-slowing device.
Standard	4-PS3-2, 4-PS3-4, 3-5-ETS1-1 — Apply scientific ideas to design a device; define criteria and constraints (4-PS3-4, 3-5-ETS1-1).
Pacing	Unit 2, week 2, lesson 2 of 2
Time	About 50 minutes
Materials	Ice cubes, cups, insulating materials, timer, workbook Lesson 2.4
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Pot handles.
Investigate	15	Ice Cube Keeper challenge.
Explain	10	Model the design process. Discuss the control cup.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook. The Weekly Check comes next.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking insulators make things warm instead of slowing heat.

SUPPORT

- Material sort with real objects.

EXTEND

- Design a lunch box that keeps food cold.

STUDENT LESSON 2.4 · WITH ANSWERS

Lesson 2.4: Conductors and Insulators

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.4 in this book with your teacher.
- [ ] Do the Investigate activity: Ice Cube Keeper.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 2.4, Part A.
- [ ] Complete the rest of Workbook Lesson 2.4 (Parts B–E).

I WONDER

Why do pot handles often have plastic or wooden grips?

Some materials let heat move through them easily. They are conductors. Most metals, such as copper, aluminum, and iron, are good conductors. That's why pots and pans are made of metal: heat moves quickly from the burner to the food.

Other materials slow heat down. They are insulators. Wood, plastic, cloth, foam, and air trapped in tiny spaces are good insulators. A plastic pot handle keeps your hand from getting burned. A winter coat traps warm air near your body. A cooler keeps ice cold by slowing heat from coming in.

Good conductors	Good insulators
copper, aluminum, iron, steel	wood, plastic, rubber, foam
used to move heat quickly: pots, pans, radiators	used to slow heat: pot handles, coats, coolers, oven mitts

EXPLAIN IT · USE THE ENGINEERING DESIGN PROCESS

Problem: keep an ice cube from melting for as long as possible.

1. Ask: What are the criteria? The ice must stay frozen as long as possible, and the holder must fit in a cup.
2. Plan: Choose insulators such as foil-covered foam, cloth, or cotton balls.
3. Build and test: Time how long the ice lasts compared with an ice cube with no holder.
4. Improve: Change one thing and test again.

Explanation: Engineers ask, plan, build, test, and improve.

INVESTIGATE · ICE CUBE KEEPER

Which design keeps an ice cube frozen longest?

You need: three ice cubes of the same size; three small cups; insulating materials: cloth, cotton balls, foam, newspaper, foil; a timer.

1 Leave one ice cube in a plain cup (your control).

2 Design and build two holders using different insulators.

3 Start the timer and check every 10 minutes.

4 Record when each ice cube is completely melted.

Record: a sketch of each design, your data table, and a sentence naming the best insulator.

★ KEY IDEA

Conductors let heat move easily; insulators slow heat. Engineers choose materials to control the flow of heat.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. Is copper a conductor or an insulator?

Answer: a conductor

2. Why are oven mitts made of thick cloth?

Answer: Cloth is an insulator that slows heat from reaching your hands.

3. What are the steps of the design process?

Answer: ask, plan, build and test, improve

Workbook Answer Key · Lesson 2.4

#	Question	Answer
1	Is copper a conductor or an insulator?	a conductor
2	Why are oven mitts made of thick cloth?	Cloth is an insulator that slows heat from reaching your hands.
3	What are the steps of the design process?	ask, plan, build and test, improve
4	A material that slows heat	insulator
5	What a design must do to succeed	criteria
6	a metal spoon	conductor
7	a wool sweater	insulator

#	Question	Answer
8	a foam cooler	insulator
9	an aluminum pan	conductor
10	a rubber pot holder	insulator
11	an iron skillet	conductor
12	Sketch your best Ice Cube Keeper and label the materials.	Labeled sketch with insulating materials.
13	How would you improve your design if you tested it again?	Answers vary; one specific change.
14	Why does a cooler keep drinks cold on a hot day?	Its insulating walls slow heat from the warm air moving into the cold drinks.

WEEKLY CHECK · ANSWER KEY
After Lesson 2.4

About 15 minutes

Weekly Check 2: How Heat Moves — Answer Key

Answers

#	Question	Answer
1	Heat moving through touching objects	conduction
2	Heat moving by flowing air or water	convection
3	How the sun's heat reaches Earth	radiation
4	Conductor or insulator: a copper pot	conductor
5	Conductor or insulator: a foam cooler	insulator
6	Why are pot handles often made of plastic or wood?	They are insulators, so heat doesn't reach your hand.

LESSON 2.5 · TEACHER PLAN
Week 3 · Lesson 1 of 2
Unit 2: Energy, Heat, and Light

Lesson 2.5: Light Travels and How We See

Student goal	I can explain that light travels in straight lines and that we see objects when light reflects off them into our eyes.
Objective	Students model how light travels and how we see.
Standard	4-PS4-2 — Develop a model to describe that light reflecting from objects and entering the eye allows objects to be seen (4-PS4-2).
Pacing	Unit 2, week 3, lesson 1 of 2
Time	About 50 minutes
Materials	Flashlights, small toys, rulers, workbook Lesson 2.5
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Book in a closet.
Investigate	15	Shadow Lengths.
Explain	10	Draw the light-path model together. Discuss why shadows form.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook lesson pages: Warm-Up through Exit Ticket.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Believing eyes send out rays to see.

SUPPORT

- Act out the light path with string.

EXTEND

- Track your outdoor shadow at three times of day.

STUDENT LESSON 2.5 · WITH ANSWERS

Lesson 2.5: Light Travels and How We See

STUDENT'S ASSIGNMENT

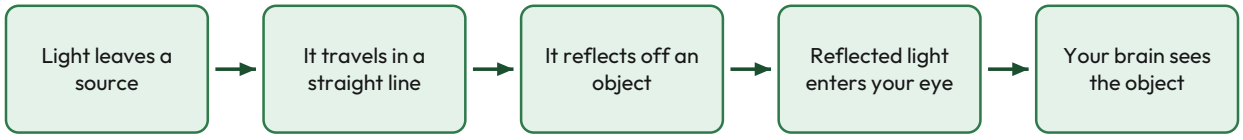
- [ ] Read Lesson 2.5 in this book with your teacher.
- [ ] Do the Investigate activity: Shadow Lengths.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 2.5, Part A.
- [ ] Complete the rest of Workbook Lesson 2.5 (Parts B–E).

I WONDER

You can see a book in a lit room, but not in a completely dark closet. What is missing in the closet?

A light source gives off its own light. The sun, a lamp, a fire, and a firefly are light sources. Most things, like a book or your hand, do not make light. We see them only because light reflects, or bounces, off them.

Light travels in straight lines from its source. When light hits an object, some of it bounces off. If that reflected light travels into your eye through your pupil, your eye sends a message to your brain, and you see the object. In a dark closet there is no light to reflect, so you can't see anything.



Because light travels in straight lines, it can't bend around an object that blocks it. This makes a shadow. A shadow is longer when the light comes from low on the side, like the sun in the early morning, and shorter when the light is high overhead, like the sun at noon.

INVESTIGATE · SHADOW LENGTHS

How does the position of a light change a shadow?

You need: a flashlight; a small toy or a pencil standing in clay; a ruler; a dark room.

1. Stand the toy on a table and shine the flashlight from straight above. Measure the shadow.
2. Shine it from a medium angle. Measure the shadow.
3. Shine it from low on the side. Measure the shadow.

4 Keep the flashlight the same distance away each time.

Record: a table of light position and shadow length, and a model drawing with straight-line arrows.

! WATCH OUT

Your eyes don't send out anything to see. Light must travel into your eyes from the object.

★ KEY IDEA

Light travels in straight lines. We see an object when light reflects off it and enters our eyes. Blocked light makes shadows.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. Name two light sources.

Answer: e.g. the sun, a lamp, a fire, a flashlight

2. Why can't you see a book in a dark closet?

Answer: There is no light to reflect off the book into your eyes.

3. When is your shadow longest: morning or noon?

Answer: morning (or evening)

Workbook Answer Key · Lesson 2.5

#	Question	Answer
1	Name two light sources.	e.g. the sun, a lamp, a fire, a flashlight
2	Why can't you see a book in a dark closet?	There is no light to reflect off the book into your eyes.
3	When is your shadow longest: morning or noon?	morning (or evening)
4	To bounce off a surface	reflect
5	Something that gives off its own light	light source
6	Draw a model showing how you see a ball in a sunny yard. Use arrows for the path of light from the sun to the ball to your eye.	Arrows from sun to ball, then ball to eye.
7	Use your Shadow Lengths data. How did the light's position change the shadow? Why?	Lower light made a longer shadow because light travels in straight lines past the object.

#	Question	Answer
8	Explain how you see the words on this page.	Light from a source reflects off the page into my eyes, and my brain makes sense of it.

LESSON 2.6 · TEACHER PLAN
Week 3 · Lesson 2 of 2
Unit 2: Energy, Heat, and Light

Lesson 2.6: Reflection, Absorption, and Transparency

Student goal	I can describe how light reflects, passes through, or is absorbed by different materials.
Objective	Students classify materials by how they interact with light.
Standard	4-PS4-2 — Develop models of how light interacts with objects (4-PS4-2).
Pacing	Unit 2, week 3, lesson 2 of 2
Time	About 50 minutes
Materials	Flashlights, mirrors, test materials, workbook Lesson 2.6
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Window vs wall.
Investigate	15	Light Detective.
Explain	10	Chart transparent, translucent, opaque. Demonstrate the mirror angle.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook. The Weekly Check comes next.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking opaque objects absorb all light.

SUPPORT

- Real-object sort.

EXTEND

- Make a mirror maze to bounce a flashlight beam around a corner.

STUDENT LESSON 2.6 · WITH ANSWERS

Lesson 2.6: Reflection, Absorption, and Transparency

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.6 in this book with your teacher.
- [ ] Do the Investigate activity: Light Detective.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 2.6, Part A.
- [ ] Complete the rest of Workbook Lesson 2.6 (Parts B–E).

I WONDER

Why can you see through a window but not through a wall?

When light hits a material, three things can happen. It can reflect (bounce off), it can pass through, or it can be absorbed (taken in). Most materials do some of each.

Kind of material	What light does	Examples
transparent	passes through almost completely	clear glass, clean water, plastic wrap
translucent	some passes through, scattered	wax paper, frosted glass, a thin sheet
opaque	is blocked: reflected or absorbed	wood, metal, cardboard, your hand

Smooth, shiny surfaces like mirrors reflect light in a neat way, so you can see an image. Light hits a mirror and bounces off at the same angle it came in, like a ball bouncing off a wall. Rough or dull surfaces scatter light in many directions.

Dark materials absorb more light than light-colored ones. Absorbed light turns into heat. That's why a black T-shirt feels hotter in the sun than a white one.

INVESTIGATE · LIGHT DETECTIVE

Which materials let light through?

You need: a flashlight; materials: clear plastic, wax paper, foil, cardboard, tissue paper, a mirror, a glass of water.

1. Predict whether each material is transparent, translucent, or opaque.
2. Shine the flashlight through each material onto a wall.
3. Record what you see on the wall: bright light, dim blurry light, or no light.

4 Shine the light at the mirror and find where the reflection lands.

Record: a three-column chart: Material | Prediction | Result.

★ KEY IDEA

Light can reflect, pass through, or be absorbed. Transparent materials let light through, translucent ones let some through, and opaque ones block it.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. Is wax paper transparent, translucent, or opaque?

Answer: translucent

2. What happens to light that hits a mirror?

Answer: It reflects (bounces off).

3. Why does a black shirt feel hotter in the sun?

Answer: Dark materials absorb more light, which turns into heat.

Workbook Answer Key · Lesson 2.6

#	Question	Answer
1	Is wax paper transparent, translucent, or opaque?	translucent
2	What happens to light that hits a mirror?	It reflects (bounces off).
3	Why does a black shirt feel hotter in the sun?	Dark materials absorb more light, which turns into heat.
4	Blocks light	opaque
5	Lets all light pass through	transparent
6	a car window	transparent
7	a wooden door	opaque
8	a lampshade	translucent
9	a cereal box	opaque
10	a clear water bottle	transparent
11	tissue paper	translucent
12	Which material would you choose for a bathroom window that lets in light but keeps privacy? Why?	Translucent (frosted) glass, because it lets light in but makes things look blurry.
13	Name the three things light can do when it hits a material.	reflect, pass through, or be absorbed

WEEKLY CHECK · ANSWER KEY
After Lesson 2.6

About 15 minutes

Weekly Check 3: Light — Answer Key

Answers

#	Question	Answer
1	Something that gives off its own light	a light source
2	Lets some light through, but blurry	translucent
3	Blocks light	opaque
4	When is a shadow longest: morning or noon?	morning
5	What bounces light to make an image you can see?	a mirror
6	Explain how you see a tree outside.	Light from the sun reflects off the tree and enters my eyes.

LESSON 2.7 · TEACHER PLAN
Week 4 · Lesson 1 of 2
Unit 2: Energy, Heat, and Light

Lesson 2.7: Refraction and Color

Student goal	I can explain that light bends when it passes into a new material and that white light is made of many colors.
Objective	Students explain refraction and color.
Standard	4-PS4-2 — Develop models of light interacting with materials (4-PS4-2).
Pacing	Unit 2, week 4, lesson 1 of 2
Time	About 50 minutes
Materials	Glasses of water, white paper, flashlight, workbook Lesson 2.7
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Broken pencil.
Investigate	15	Make a Spectrum.
Explain	10	Demonstrate refraction with a pencil. Explain reflected color.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook lesson pages: Warm-Up through Exit Ticket.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking objects “have” color instead of reflecting it.

SUPPORT

- Colored filters to look through.

EXTEND

- Look at a red object through a blue filter and explain what you see.

STUDENT LESSON 2.7 · WITH ANSWERS

Lesson 2.7: Refraction and Color

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.7 in this book with your teacher.
- [ ] Do the Investigate activity: Make a Spectrum.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 2.7, Part A.
- [ ] Complete the rest of Workbook Lesson 2.7 (Parts B–D).

I WONDER

Put a pencil in a glass of water and look from the side. Why does it look broken?

Light travels at different speeds in different materials. When it passes from air into water or glass at an angle, it changes speed and bends. This bending is called refraction. It makes a pencil in water look broken and makes a pool look shallower than it really is.

A lens is a curved piece of clear glass or plastic that bends light on purpose. Eyeglasses, magnifying glasses, cameras, and the lens inside your own eye all use refraction to focus light.

Sunlight looks white, but it is really a mix of colors. A prism refracts each color a slightly different amount, spreading white light into a spectrum: red, orange, yellow, green, blue, and violet. A rainbow forms when raindrops act like tiny prisms.

Why does a red apple look red? The apple absorbs most colors of light and reflects mostly red light into your eyes. A white object reflects all colors; a black object absorbs almost all of them.

INVESTIGATE · MAKE A SPECTRUM

Can you split white light into colors?

You need: a clear glass of water; a sheet of white paper; a sunny window or a bright flashlight.

1. Set the glass of water on a sunny windowsill.
2. Hold the white paper on the floor or table where light passes through the water.
3. Move the paper until you see colors.
4. Record the order of the colors you see.

Record: a colored drawing of your spectrum with the colors in order.

★ KEY IDEA

Refraction is the bending of light when it passes into a new material. Prisms and raindrops separate white light into a spectrum. We see an object's color from the light it reflects.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. Why does a pencil in water look broken?

Answer: Light bends (refracts) when it passes from water into air.

2. What colors are in white light?

Answer: red, orange, yellow, green, blue, and violet

3. Why does a banana look yellow?

Answer: It reflects mostly yellow light and absorbs the other colors.

Workbook Answer Key · Lesson 2.7

#	Question	Answer
1	Why does a pencil in water look broken?	Light bends (refracts) when it passes from water into air.
2	What colors are in white light?	red, orange, yellow, green, blue, and violet
3	Why does a banana look yellow?	It reflects mostly yellow light and absorbs the other colors.
4	The bending of light	refraction
5	The band of colors in white light	spectrum
6	Name a tool that uses a lens.	e.g. eyeglasses, a magnifying glass, a camera
7	What acts like a prism to make a rainbow?	raindrops
8	Which color does a green leaf reflect?	green
9	Which reflects the most colors: a white shirt or a black shirt?	a white shirt
10	Explain how a rainbow forms.	Sunlight passes into raindrops, which refract and separate white light into its colors.

LESSON 2.8 · TEACHER PLAN
Week 4 · Lesson 2 of 2
Unit 2: Energy, Heat, and Light

Lesson 2.8: Light Investigation: Claim, Evidence, and Reasoning

Student goal	I can plan an investigation about light and explain my results with a claim, evidence, and reasoning.
Objective	Students build a periscope and write a CER explanation.
Standard	4-PS4-2, Science and Engineering Practices 3, 7 — Plan and carry out investigations; construct explanations (SEP 3, 6, 7).
Pacing	Unit 2, week 4, lesson 2 of 2
Time	About 50 minutes
Materials	Cartons, small mirrors, tape, scissors, workbook Lesson 2.8, Unit 2 Review
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Submarine sailors.
Investigate	15	Build a Periscope.
Explain	10	Model a CER paragraph. Trace the light path.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook; then the Unit 2 Review.
Exit Ticket	3	Next: Unit 2 Test.

COMMON MISCONCEPTIONS

- Writing evidence without reasoning.

SUPPORT

- CER sentence frames.

EXTEND

- Make your periscope longer and test whether it still works.

STUDENT LESSON 2.8 · WITH ANSWERS

Lesson 2.8: Light Investigation: Claim, Evidence, and Reasoning

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.8 in this book with your teacher.
- [ ] Do the Investigate activity: Build a Periscope.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 2.8, Part A.
- [ ] Complete the rest of Workbook Lesson 2.8 (Parts B–E).

I WONDER

Submarine sailors look above the water without coming up. How might they do it?

Scientists answer questions by planning investigations, collecting evidence, and explaining their results. A strong explanation has three parts: a claim, evidence, and reasoning (CER).

Part	What it does	Example
claim	answers the question	A periscope lets you see over a wall.
evidence	data you collected	I saw the clock across the room while I was hiding behind the couch.
reasoning	the science idea that connects them	Light travels in straight lines and reflects off mirrors at an angle, so the mirrors carried the clock's light down the tube to my eye.

A periscope uses two mirrors set at an angle. Light from an object enters the top opening, reflects off the top mirror straight down the tube, reflects off the bottom mirror, and travels out into your eye.

INVESTIGATE · BUILD A PERISCOPE

Can mirrors let you see over a wall?

You need: a long cardboard box (like a juice or milk carton, cleaned); two small mirrors; tape; scissors (with an adult).

1. Cut a square window near the top of one side and near the bottom of the opposite side.
2. Tape one mirror inside the top, tilted to face down the tube; tape the other inside the bottom, tilted to face the bottom window.
3. Look through the bottom window and adjust the mirrors until you see out the top.

4 Hide below a table or couch and test what you can see.

Record: a labeled diagram with arrows showing the light path, and a CER paragraph.

Safety: An adult should help with cutting. Handle mirrors carefully.

★ KEY IDEA

Scientists explain results with a claim, evidence, and reasoning. A periscope works because light travels in straight lines and reflects off mirrors.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. What are the three parts of a scientific explanation?

Answer: claim, evidence, and reasoning

2. How many mirrors does a periscope use?

Answer: two

3. What science idea explains how a periscope works?

Answer: Light travels in straight lines and reflects off mirrors.

Workbook Answer Key · Lesson 2.8

#	Question	Answer
1	What are the three parts of a scientific explanation?	claim, evidence, and reasoning
2	How many mirrors does a periscope use?	two
3	What science idea explains how a periscope works?	Light travels in straight lines and reflects off mirrors.
4	A statement that answers a question	claim
5	Data that supports a claim	evidence
6	Draw your periscope from the side. Show both mirrors and draw arrows for the path of light into your eye.	Two angled mirrors with a light path from object to top mirror to bottom mirror to eye.
7	Write a CER paragraph: Can a periscope let you see over a wall?	Claim: yes. Evidence: what I saw while hidden. Reasoning: light travels in straight lines and reflects off the mirrors into my eye.
8	Why do you need the reasoning part of CER, not just the evidence?	Reasoning explains the science idea that connects the evidence to the claim.

UNIT REVIEW · ANSWER KEY
Study before the unit test

About 40 minutes

Unit 2 Review — Answer Key

Homework or a study session before the test.

Answers

#	Question	Answer
1	temperature	how hot or cold something is
2	conduction	heat through touching
3	insulator	slows heat
4	reflect	bounce off
5	refraction	bending of light
6	opaque	blocks light
7	Draw arrows to show how heat moves from a stove burner, through a pan, and into food.	Arrows from burner to pan to food (conduction).
8	Draw a model that shows how you see a book in a lit room.	Arrows from the light source to the book, then from the book to the eye.

UNIT TEST · ANSWER KEY
At the end of the unit

About 45 minutes

Unit 2 Test — Answer Key

Answers

#	Question	Answer
1	Which marble has the most motion energy? A. a still marble B. a slow marble C. a fast marble	C. a fast marble
2	Name two forms of energy.	any two of motion, sound, light, heat, electrical
3	Water boils at… (°C)	100°C
4	Heat always flows… A. from cooler to warmer B. from warmer to cooler C. only through metal	B. from warmer to cooler
5	Warm air rising from a heater is an example of…	convection
6	Feeling heat from a campfire is an example of…	radiation
7	A spoon getting hot in soup is an example of…	conduction
8	Which is the best insulator? A. an aluminum pan B. a wool mitten C. a copper wire	B. a wool mitten
9	Why can't you see in a completely dark room?	There is no light to reflect off objects into your eyes.
10	Wax paper is… A. transparent B. translucent C. opaque	B. translucent
11	Why does a pencil look broken in water?	Light refracts (bends) between water and air.
12	Why does a red apple look red?	It reflects red light and absorbs the other colors.
13	You want to keep a cup of hot cocoa warm for a long time. Design a way to do it. Name the materials you would use, explain how heat moves, and explain why your design works.	Rubric: names insulating materials (1); describes heat flowing from warm cocoa to cooler air (1); names a way heat moves, such as conduction or convection (1); explains how the insulator slows it (1).

Scoring

Score	What it means / next step
14–16 (85%+)	Mastered.
12–13	Proficient; review missed skills.
Below 12	Reteach and retest.

Grade 4 Science

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INSIDE THIS YEAR

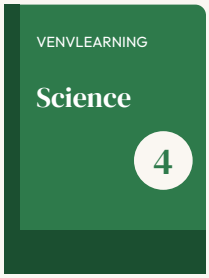
01 The Human Body

02 Energy, Heat, and Light

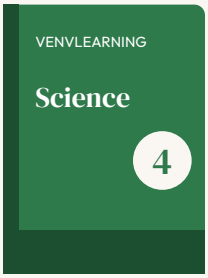
IN THIS BOOK

- Week-by-week pacing guide
- Guided lessons with timing and teaching moves
- Misconceptions, supports, and extensions
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